

Non-Functional Requirements (HVPN-NFR-NNN)

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Revision: 2 Last modified: 2026-07-04T12:00:00Z

Rev 2 (2026-07-04, independent gap-analysis pass).

Added five NFR classes that were absent or thin against enterprise-production-readiness criteria: secrets management/rotation (NFR-411), WireGuard/device-key rotation cadence + triggers (NFR-412), control-plane API rate limiting (NFR-413), data-plane DDoS/UDP-flood/amplification resilience (NFR-414) — a *different* concern from NFR-413 (control-plane request throttling), audit-retention default (NFR-309), and a new **Operability & Compatibility** band (NFR-700...) covering zero-downtime rolling deploys, Postgres schema-migration backward-compatibility windows, protobuf wire-compatibility guarantees, and an infra-sizing/cost model. No existing NFR was weakened; §9 traceability and the family diagram are extended to match.

Document role. This is the Volume 1 (Product & Requirements) nano-detail spec enumerating every HelixVPN **non-functional requirement (NFR)** — performance, scale, availability/HA, privacy, security, resource, and portability — each numbered HVPN-NFR-NNN. It is the quantified counterpart to the *seven non-negotiable principles* and the *Mullvad-parity matrix* of the Volume 1 overview (00-product-scope-and-principles.md). Where a value is a **target not yet proven on hardware** it is stated as a TARGET and marked UNVERIFIED per §11.4.6 (no-guessing) — a number that only a Phase-0/1 gate or a captured benchmark can confirm is never asserted as established fact here.

Document is a SPEC. It states *what quality bars the product must meet and how each is verified*; it does not build the product. Every NFR carries a measurable target plus the §11.4.169 test type that proves it, so each row is falsifiable rather than aspirational (§11.4 / §11.4.169).

Evidence base. Cross-references the Volume 1 overview (00-product-scope-and-principles.md — principles P1-P7, parity matrix F1-F17, decisions D1-D7, scope §10), the control-plane coordinator SLO (v03-control-plane/svc-coordinator.md §7 — convergence p99 < 1 s), the telemetry counts-only metric registry (v03-control-plane/svc-telemetry.md §3 — what may be measured), the no-logging invariant (v05-security/no-logging-as-code.md — the privacy NFRs as testable bars), and the transport ladder (v02-data-plane/transport-selection-ladder.md). Source research is cited by the same ids the overview uses ([04_ARCH §N], [04_P0], [04_P1]).

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1. How to read an NFR row

Every NFR is one row with five fixed fields, so each is independently testable (§11.4 — a quality bar with no measurement is a PASS-bluff at the requirements layer):

Field	Meaning
ID	HVPN-NFR-NNN, stable, append-only — the cross-doc reference key (§11.4.54 discipline applied to NFRs).
Statement	the qualitative requirement (one sentence).
Target	

Field	Meaning
	the measurable bar (number + unit + percentile/condition). TARGET = not yet proven on hardware (UNVERIFIED §11.4.6).
Verify by	the §11.4.169 test type whose captured evidence proves it (unit / integration / e2e / performance / soak-stress / chaos / security / memory / benchmarking).
Priority	P0 (Phase-0 gate / make-or-break), MVP (Phase-1 DoD), P2/P3 (later phase), ALWAYS (every phase).

Priority bands map to the §10 scope spine of the overview: a P0 NFR is a make-or-break Phase-0 exit gate; an MVP NFR is part of the 8-criteria Definition-of-Done; a P2/P3 NFR binds when its phase opens.

Honest-number rule (§11.4.6). Several targets below originate from the refined research as *design targets* (e.g. the iOS NE memory ceiling, the $\geq 80\%$ -bare-link throughput gate). They are reproduced as TARGETs and marked UNVERIFIED — the Phase-0 gate or the named benchmark is what converts a TARGET into an established fact. No target is restated here as if already measured.

2. Performance NFRs (HVPN-NFR-001...)

Performance is dominated by two budgets: **datapath throughput/latency** (the Rust data plane, doc 01) and **control-plane convergence** (the coordinator, svc-coordinator.md).

ID	Statement	Target
HVPN-NFR-001	Plain-UDP WireGuard datapath sustains near-bare-link throughput client→gateway→connector-LAN.	$\geq 80\%$ of bare-link throughput (Phase-0 gate G1) — TARGET UNVERIFIED
HVPN-NFR-002	MASQUE/QUIC obfuscated datapath retains usable throughput through a DPI UDP block.	$\geq 50\%$ of plain-UDP throughput (Phase-0 gate G2) — TARGET UNVERIFIED
HVPN-NFR-003	Control-plane convergence: a policy/topology change reaches every affected agent's WatchNetworkMap stream.	p99 < 1 s event→delta-on-wire (helix_reconcile_seconds)
HVPN-NFR-004	Device revoke is enforced (peer-removal delta on wire + stream force-close).	< 1 s revoke-event→sub.cancel()

ID	Statement	Target
HVPN-NFR-005	WireGuard handshake completes within a bounded latency on a healthy link.	(helix_revoke_enforce_seconds p99) handshake-complete TARGET ≤ 1 s typical; rung success = WG handshake, not carrier connect — UNVERIFIED exact bound
HVPN-NFR-006	Transport-ladder escalation to a working rung is bounded per attempt (no unbounded hang).	per-rung budget enforced; cursor advances monotonically (L-I8); total $\leq \Sigma$ rung budgets — UNVERIFIED calibrated value
HVPN-NFR-007	Datapath added latency (encryption + obfuscation overhead) stays within an obfuscation-aware budget.	plain-UDP overhead TARGET near-zero; MASQUE adds QUIC/H3 framing cost, bounded — UNVERIFIED measured value
HVPN-NFR-008	MTU is chosen so WG-in-transport never fragments on a 1500-byte path.	WG MTU 1420 plain; transport-specific reductions for MASQUE/Shadowsocks documented per rung
HVPN-NFR-009	Admin-perceived "policy edit reflected" end-to-end (REST→compile→activate→bus→delta).	< 1 s as the MVP DoD-5 criterion (superset of NFR-003, per-hop timers)

Note on NFR-003 vs NFR-009 (§11.4.6). NFR-003 measures the coordinator's event-receive → stream.Send segment only; NFR-009 is the wider admin-edit path the DoD asserts and is a *superset* — they are tracked as distinct series (svc-coordinator.md §7.1, svc-telemetry.md §6.1), not conflated.

3. Scale NFRs (HVPN-NFR-100...)

Scale targets are stated as MVP single-node bars plus Phase-2 fleet bars. Where the MVP number is a sizing target not yet load-proven it is UNVERIFIED.

ID	Statement	Target	Verify by	Priority
HVPN-NFR-100	A single self-hosted gateway pod serves a homelab-scale fleet of	TARGET ≥ 10 k concurrent WatchNetworkMap streams per coordinator (the soak-test figure) — UNVERIFIED	soak-stress (24 h, N=10 k streams)	MVP

ID	Statement	Target	Verify by	Priority
HVPN-NFR-101	concurrent agents. Coordinator memory stays bounded under sustained stream load (no leak).	process_resident_memory_bytes slope ≈ 0 over 24 h @ 10 k streams	soak-stress / memory	MVP
HVPN-NFR-102	The minimal-affected-set fan-out keeps per-event work small regardless of tenant size.	helix_fanout_affected_nodes $\ll$ tenant size for a small change	unit + performance	MVP
HVPN-NFR-103	One gateway supports many joined private networks (connectors) per tenant.	TARGET connectors/tenant bounded only by overlay-IP space (ULA /48, D4) — UNVERIFIED concrete ceiling	integration (multi-connector overlay)	MVP
HVPN-NFR-104	One user reaches N joined networks (the multi-network differentiator X1).	1 user $\rightarrow$ N nets, N bounded by policy + IPAM, not architecture	e2e (multi-network reach)	MVP
HVPN-NFR-105	Reconnect storm (mass gateway flap) is absorbed without deadlock or unbounded buffering.	10 k simultaneous reconnects: bounded send queues, no deadlock, drops counted not buffered	stress (reconnect-storm)	MVP
HVPN-NFR-106	Per-tenant isolation does not let one tenant's churn block another.	per-tenant tenantState sharding; one tenant's lock never held across another	concurrency / race-deadlock	MVP

ID	Statement	Target	Verify by	Priority
HVPN-NFR-107	Control plane scales horizontally to stateless multi-replica coordinators.	coordinators stateless (graph rebuilt from Postgres+events on boot); N replicas, no sticky state	integration (multi-replica)	P2
HVPN-NFR-108	Event backbone scales from single-node Redis Streams to multi-region NATS JetStream.	bus interface (D3) transport-agnostic; swap is mechanical, not a redesign	integration (bus swap)	P2

4. Availability & HA NFRs (HVPN-NFR-200...)

The load-bearing availability property is **fail-static** (P1): a control-plane outage is never a connectivity outage.

ID	Statement	Target	Verify by	Pri
HVPN-NFR-200	Existing tunnels keep forwarding while the control plane is down (fail-static, P1).	0 tunnel drops on control-plane outage; edge forwards from cached verdict map	chaos (kill control plane mid-traffic)	ALV
HVPN-NFR-201	A Postgres/Redis blip does not restart the gateway process.	liveness probe dependency-free; readiness gates new traffic only, never kills serving streams	chaos (DB/Redis blip)	MV
HVPN-NFR-202	Presence backend (Redis) loss degrades gracefully, never errors a tunnel (fail-static).	IsOnline→ErrPresenceUnavailable; coordinator assumes reachable-via-relay degraded; no tunnel error	chaos (drop Redis mid-stream)	MV
HVPN-NFR-203	A crashed coordinator consumer loses	XAutoClaim reclaim of stalled PEL entries; 0 lost events across a kill	chaos (kill consumer mid-apply)	MV

ID	Statement	Target	Verify by	Pri
HVPN-NFR-204	no events (no-work-loss). A coordinator restart is transparent to connected agents (resume by known_version).	agents resume from known_version against the rebuilt-from-Postgres graph; snapshot fallback on ring gap	chaos / integration (restart mid-stream)	MV
HVPN-NFR-205	Multi-region HA: stateless coordinators + Patroni Postgres + NATS JetStream survive a region loss.	TARGET region-failover within an RTO/RPO budget — UNVERIFIED (Phase-2 DR runbook owns the number)	chaos (region-failover drill)	P2
HVPN-NFR-206	Gateway failover re-points agents to a healthy gateway endpoint.	gateway.failover event → every tenant agent gets an updated endpoint within the convergence SLO	integration / chaos	P2
HVPN-NFR-207	Poison events never spin the consumer; they route to a DLQ after a delivery-count ceiling.	delivery-count ≥ cap → DLQ + alert; no infinite retry	chaos / integration (DLQ)	MV

Availability SLO numbers are deferred to the DR doc (§11.4.6). The concrete uptime %, RTO, and RPO budgets live in v06-deploy/disaster-recovery.md (which closes ledger gap G1); reproducing a specific "99.9 %" here would be a fabricated number — NFR-205 names the budget's *owner*, not a guessed value.

5. Privacy NFRs (HVPN-NFR-300...)

Privacy is the product's reason to exist; these NFRs make the no-logging guarantee **testable** rather than promissory. They are bound by no-logging-as-code.md — what may be measured is constrained by what may exist.

ID	Statement	Target	Verify by	Priority
HVPN-NFR-300	No durable connection/traffic/packet/flow/session table exists anywhere in the schema (P7, S6).	schema-lint exit 0 against migrations AND the live information_schema	security (schema-lint, pre-build + post-deploy runtime signature)	ALWAYS
HVPN-NFR-301	The schema-lint is not a tautology (it actually catches a planted flow table).	planted flows(src,dst,bytes,ts) migration → lint FAILs ; removed → PASSes (paired §1.1)	meta-test (§1.1 golden-bad/good)	ALWAYS
HVPN-NFR-302	The event bus carries no traffic shape (C3 on the bus).	zero src/dst/bytes/dns fields in any event payload struct; payload-lint green	security (payload-lint, build + publish-time)	MVP
HVPN-NFR-303	Audit records control actions only, never traffic; the meta jsonb cannot smuggle a flow.	closed action vocabulary; meta key matched against forbidden regex → ErrAuditMetaShape	unit + integration (audit meta guard)	MVP
HVPN-NFR-304	Live presence is ephemeral (Redis TTL), never persisted to Postgres.	presence:{tenant}:{device} TTL = 3× heartbeat; no durable session row; loss of Redis loses no durable state	integration (presence TTL/expiry)	MVP
HVPN-NFR-305	The one durable presence derivative (last_seen_at) cannot reconstruct a session timeline.	coarsened to ≥ 5 min (LAST_SEEN_COARSEN), carries no destination	unit + integration	MVP
HVPN-NFR-306	/metrics exposes aggregate counters only — no per-	every collector's label set $\subseteq$ allow-list; no tenant_id/device_id/*_ip label	unit (label-cardinality audit)	MVP

ID	Statement	Target	Verify by	Priority
	user, per-flow, per-destination, or per-tenant-label series.			
HVPN-NFR-307	Anonymous identity is supported (no email, no SSO required).	a tenant mints device enroll tokens with <code>users.email = NULL, oidc_sub = NULL</code> ; no reverse-link to a human	integration (anonymous enroll)	MVP
HVPN-NFR-308	The no-logging guarantee holds as a runtime signature, not a source grep (§11.4.108).	schema-lint green against the deployed DB, asserted post-deploy	e2e Challenge (deployed-DB lint)	MVP
HVPN-NFR-309	Audit (control-action) retention follows an explicit, operator-set policy — never a silent default that either grows unbounded or auto-prunes accountability evidence.	default = no auto-prune (D-AC-3); operator MAY set a retention period per their compliance regime; retention config is itself an audited control action	integration (retention-policy set/enforce)	MVP

Honest boundary (§11.4.6). These NFRs guarantee *durable-store + bus + audit* absence of traffic data. They do **not** claim a hot-compromised running gateway holds nothing in RAM (live presence keys + aggregate counters are the residual surface T5 in *no-logging-as-code.md* §14). The privacy NFR set is "no durable user↔destination correlation ever exists," never "an attacker with root on a live box sees nothing."

6. Security NFRs (HVPN-NFR-400...)

ID	Statement	Target	Verify by	Priority
HVPN-NFR-400	WireGuard's audited crypto core is never forked; obfuscation is a layer <i>under</i> it (P2).	WG Noise IK / Curve25519 / ChaCha20-Poly1305 untouched; CI lint forbids edits to WG crypto primitives	security + meta-test	ALWAYS
HVPN-NFR-401	Edges are outbound-only; no private network needs an inbound port (P3).	only the gateway listens publicly; connectors/clients dial out	security (no-inbound assertion)	ALWAYS
HVPN-NFR-402	Default-deny, need-to-know: a node only ever learns peers a compiled rule grants (C4).	a node with no VisibleTo entry sees zero peers; deny-unauthorized peer never appears in any snapshot	e2e + meta-test (skip-filter mutation FAILs)	MVP
HVPN-NFR-403	Device identity is mTLS with short-lived certs and instant revocation.	short-lived device cert; revoke enforced < 1 s (NFR-004); impersonation (device_id≠cert) rejected	security	MVP
HVPN-NFR-404	Kill-switch prevents any leak if the tunnel drops.	0 packets off-tunnel on tunnel drop (OS firewall state machine)	security / e2e (leak test on drop)	MVP
HVPN-NFR-405	DNS-leak protection forces tunnel DNS and blocks plaintext off-tunnel :53.	0 plaintext DNS off-tunnel	security / e2e (DNS-leak test)	MVP
HVPN-NFR-406	The device private WG key	key generated on-device at enrollment; only	integration + security	MVP

ID	Statement	Target	Verify by	Priority
	never leaves the device.	the public key is registered (C6)		
HVPN-NFR-407	Post-quantum handshake is hybrid (PQ pre-shared layer), never PQ-only.	ML-KEM/ FIPS-203 PSK layered on WG; classical handshake retained — hybrid-never-PQ-only	security + integration	P2
HVPN-NFR-408	Multi-tenant isolation is enforced at the database (RLS), not only in app code.	FORCE ROW LEVEL SECURITY; tenant A cannot read tenant B even with a crafted query as helix_app	security / integration (RLS)	MVP
HVPN-NFR-409	Every change crosses the mandatory anti-bluff test gauntlet before release.	each §11.4.169 type present + paired §1.1 mutation that flips its gate RED	meta-test (per-gate mutation)	ALWAYS
HVPN-NFR-410	Censorship-evasion: the ladder reaches a working obfuscated transport under DPI/UDP-block.	escalation reaches MASQUE (or a later rung) and completes a WG handshake under a DPI sim	e2e (DPI-sim escalation)	MVP
HVPN-NFR-411	Control-plane secrets (Postgres/Redis credentials, tenant CA root/ KMS key material, OIDC client secrets) are never stored in plaintext in the image or repo, and are rotatable without a rebuild.	secrets injected at runtime (env/ File/KMS reference, never baked into an image layer); a credential rotation (DB password, Redis AUTH) completes with 0 dropped in-flight control RPCs	security + integration (rotate-while-serving)	MVP
HVPN-NFR-412	Device WireGuard keys	device WG key: operator-	security + integration	MVP

ID	Statement	Target	Verify by	Priority
		configurable rotation period (TARGET default 90 days, UNVERIFIED		
	and gateway/ edge transport keys rotate on an explicit cadence and immediately on revocation — never silently indefinite.	concrete default) and immediate re-key on device.revoked; gateway edge transport key: independent rotation schedule from device keys, rotated with 0 tunnel drop for unaffected peers	(scheduled + on-revoke rotation, zero-drop capture)	
HVPN-NFR-413	The control-plane API (Gin REST + gRPC/ Connect agent surface) enforces per-principal rate limiting so no single API key/ device can exhaust coordinator or Postgres capacity.	token-bucket per API key/ device on api+coordinator; TARGET default ceiling UNVERIFIED (operator-tunable), 429/ backpressure response, no unbounded queueing (composes NFR-506)	stress (burst-request flood) + security	MVP
HVPN-NFR-414	The data-plane edge resists UDP-flood / reflection-amplification abuse of its public :443/udp (and other) listeners — a <i>distinct</i> concern from NFR-413 (that governs the authenticated	WireGuard's built-in stateless cookie-reply mechanism is enabled (bounds per-source handshake-initiation cost); the MASQUE/ QUIC listener rate-limits unmatched/pre-	DDoS test (v08-testing/ddos.md) — flood simulation with pre/post resource + amplification-ratio capture	MVP

ID	Statement	Target	Verify by	Priority
	control-plane API; this governs the unauthenticated public data-plane listener).	handshake datagrams per source IP; edge CPU/bandwidth stay bounded under a simulated flood (no amplification — reply size $\leq$ request size for any unauthenticated packet)		

7. Resource NFRs (HVPN-NFR-500...)

The single hardest constraint in the whole product is the **iOS Network Extension memory ceiling** — it shapes decision D2 (Rust core).

ID	Statement	Target	Verify by
HVPN-NFR-500	The Rust client core runs inside the iOS NEPacketTunnelProvider memory budget with headroom (make-or-break).	working-set under the iOS NE ceiling with $\geq 30\%$ headroom (Phase-0 gate G3) — the historical ~15 MB ceiling is a measured constraint, TARGET UNVERIFIED until G3 on-device	memory (on-device NE soak)
HVPN-NFR-501	Mobile client is battery-sensitive: push-don't-poll, no busy loops.	event-driven reconcile (no cron/poll); TARGET measurable battery delta vs idle — UNVERIFIED	benchmarking (battery/CPU on-device)
HVPN-NFR-502	The client core fits a small binary footprint on size-constrained platforms.	TARGET core .so/.a size budget per platform — UNVERIFIED concrete bytes	benchmarking (artifact-size check)
HVPN-NFR-503	Project procedures (build/test/tooling) never exceed 60 % of host RAM (§12.6).	host-side resource ceiling $\leq 60\%$ total RAM	benchmarking (host resource sampler §11.4.24)
HVPN-NFR-504	The control-plane process holds bounded memory		soak-stress / memory

ID	Statement	Target	Verify by
	under sustained load (no leak).	process_resident_memory_bytes slope ≈ 0 over 24 h (= NFR-101, server side)	
HVPN-NFR-505	Telemetry samplers (observers) stay lightweight (Heisenberg constraint).	a resource sampler stays under its own RSS/CPU budget (§11.4.24: < 50 MB RSS, < 5 % CPU)	benchmarking
HVPN-NFR-506	Slow-consumer streams are dropped, not buffered unboundedly (back-pressure not growth).	per-stream SEND_QUEUE_CAP bound; overflow $\rightarrow$ drop + reconnect-with-snapshot	stress / memory

NFR-500 is the program's pivot (§11.4.6). The overview names G3 as the literal decider for D2: if Rust cannot fit the iOS NE ceiling with headroom, D2 *and the whole client strategy* re-open. The ~15 MB figure is reproduced as the historical measured constraint and marked UNVERIFIED until the G3 on-device capture exists — it is never asserted here as a settled fact.

8. Portability NFRs (HVPN-NFR-600...)

HelixVPN targets **8 platforms on one shared codebase** (Rust core + Flutter UI

- per-platform tunnel shims). Portability NFRs are per-platform reachability bars, phased per §10.

ID	Statement	Target	Verify by	Priority
HVPN-NFR-600	All three apps build from one Flutter tree via the <code>runHelixApp(flavor,...)</code> entrypoint.	flavors {Access, Connector, Console}; Console omits <code>core_ffi</code> (no tunnel core)	integration (per-flavor build)	MVP
HVPN-NFR-601	The Rust core is shared byte-for-byte across Client, Connector, and Gateway edge (P5).	one helix-transport crate; wrap-on-client == unwrap-on-edge	integration (shared-core parity)	MVP
				MVP

ID	Statement	Target	Verify by	Priority
HVPN-NFR-602	MVP Access app runs on iOS, Android, and Linux.	three platforms green on the 8-criteria DoD	e2e / full-automation	
HVPN-NFR-603	Connector daemon runs headless on Linux/Windows/macOS (+ Android/embedded for appliances).	headless daemon + optional slim UI; advertise/route mode	integration	MVP
HVPN-NFR-604	Console runs responsively as Web and as a desktop Flutter build.	one Flutter build serves Web + Desktop; admin flavor, API-client only	e2e (responsive Web + desktop)	MVP
HVPN-NFR-605	Phase-2 desktop apps reach Windows (wireguard-nt+service) and macOS (NE).	Windows + macOS Access apps green	e2e	P2
HVPN-NFR-606	Phase-3 extends to HarmonyOS NEXT and Aurora OS (the biggest platform risk).	native tunnel-shim work green on both — TARGET UNVERIFIED (no incumbent precedent)	e2e (per-platform shim)	P3
HVPN-NFR-607	The Web build is honestly scoped: Console + an <i>optional</i> browser-scoped WASM MASQUE proxy, never a system VPN.	no TUN device in-browser; the WASM proxy proxies the browser's own traffic only	e2e + documentation honesty check	P3
HVPN-NFR-608	Every UI surface ships light + dark variants and passes visual-regression (§11.4.162 OpenDesign).	light+dark per component; no overlap/overlay; visual-regression green	UI / visual-regression	P3
				ALWAYS

ID	Statement	Target	Verify by	Priority
HVPN-NFR-609	Cross-platform parity: any platform-specific primitive has a per-OS equivalent or an honest gap (§11.4.81).	uname -s-dispatched equivalents or documented kernel-gap SKIP — never a silent Linux-only pass	per-OS test branches	

8a. Operability & Compatibility NFRs (HVPN-NFR-700...)

These NFRs close a gap the earlier revision left implicit: a self-hostable product that scales to a managed fleet (P6) **MUST** also be **upgradeable** and **forecastable** without a rewrite. They are the "can we run this for years, across upgrades and growth" bars — distinct from the runtime performance/scale bars above.

ID	Statement	Target	Verify by	Priority
HVPN-NFR-700	The control plane supports zero-downtime rolling upgrades: a new version deploys without dropping in-flight WatchNetworkMap streams or existing data-plane tunnels.	rolling deploy (N replicas, one at a time) completes with 0 dropped agent streams beyond the individual replica's own graceful- drain reconnect (which resumes from known_version, NFR-204); existing edge tunnels never drop (fail- static, NFR-200 holds <i>across</i>	chaos + integration (rolling- upgrade drill)	P2

ID	Statement	Target	Verify by	Priority
HVPN-NFR-701	Postgres schema migrations are backward-compatible for at least one prior minor version (expand-then-contract), so a rolling upgrade never requires simultaneous downtime of app + DB migration.	a control-plane version change too) every migration is additive-first (new column nullable/defaulted, old column dropped only in a later release after the app no longer reads it); a mixed-version fleet (N replicas on version X, N on X+1) runs correctly against one schema version	integration (mixed-version-fleet-against-one-schema test)	P2
HVPN-NFR-702	Protobuf (helix.coordinator.v1) and OpenAPI contracts are wire-compatible within a documented backward-compatibility window: new optional fields only, no field-number reuse, no required-field addition.	a buf/OpenAPI breaking-change linter runs in the codegen pipeline (v06-deploy/codegen-pipeline.md) and fails the build on a breaking change without an explicit major-version bump	unit (schema-diff lint) + meta-test (planted breaking change → lint FAILs)	MVP
HVPN-NFR-703	Infra sizing (vCPU/RAM/disk/network) scales predictably with tenant/device count, so an operator	a documented sizing formula (or reference table) ties	benchmarking (resource-vs-load curve from the soak-stress suite, v06-deploy/	P2

ID	Statement	Target	Verify by	Priority
	can forecast cost before growing.	coordinator/ Postgres/ Redis/edge resource needs to {tenants, devices, connectors} counts, validated against the NFR-100/101 soak-test figures; TARGET concrete formula UNVERIFIED until the soak-test dataset exists	observability.md sizing dashboard)	

Honest boundary (§11.4.6). NFR-703 is deliberately not a fixed dollar figure — cloud/self-host pricing varies by operator and region. The NFR requires a *sizing formula* (resource units per tenant/device), not a cost quote; a managed-SKU cost model (if D8 licensing lands a managed offering) is a commercial decision layered on top of this technical sizing curve, tracked separately from this spec set.

9. NFR → principle → verification traceability

Each NFR family traces to the overview's non-negotiable principles (P1-P7) and the parity matrix (F1-F17 / X1-X5), so no quality bar floats unanchored (§11.4.25 coverage discipline).

NFR family	Principle(s)	Parity / differentiator rows	Primary §11.4.169 test types
Performance (001-009)	P1 (fail-static keeps Go off hot path), P4 (push)	F1, F2, F7	performance, benchmarking, integration
Scale (100-108)	P4, P6 (self-host→fleet)	X3	

NFR family	Principle(s)	Parity / differentiator rows	Primary §11.4.169 test types
			soak-stress, stress, concurrency
Availability/ HA (200–207)	P1 (fail- static)	X3	chaos, integration
Privacy (300– 308)	P7 (no- logging by construction)	F14, F15	security (schema/ payload lint), meta-test
Security (400–410)	P2 (WG core), P3 (outbound- only)	F11, F12, F13, F16, F17, X2, X4	security, e2e, meta-test
Resource (500–506)	P5 (one core), P6	F1, X5	memory, benchmarking
Portability (600–609)	P5 (one core, reused)	X5	integration, e2e, UI/visual- regression
Operability & Compatibility (700–703)	P6 (self- host→fleet, no rewrite)	X3	chaos, integration, unit (schema- diff), benchmarking

[MERMAID RENDER FAILED — block #1]

10. The convergence-NFR dependency chain

NFR-003 (p99 < 1 s convergence) and NFR-004 (< 1 s revoke) are the two headline real-time bars; they depend on a chain of lower NFRs holding, and the telemetry service makes each link a falsifiable series (counts/timing only, never traffic — svc-telemetry.md §3).

[MERMAID RENDER FAILED — block #2]

The dependency order (each lower NFR is a precondition of the SLO above it):

1. **NFR-101 / NFR-506** (bounded memory + back-pressure) — without these the coordinator could not hold 10 k streams to *deliver* a sub-second delta.

2. **NFR-102** (minimal-affected-set) — a tenant-wide broadcast on every change would blow the 1 s budget; the minimal set keeps per-event work small.
3. **NFR-203 / NFR-207** (no-work-loss + DLQ) — a stuck/lost event never converges at all, so reliability is a precondition of latency.
4. **NFR-200** (fail-static) — convergence is a *coordination* property; even when it is breached, connectivity (the edge datapath) must not be.

§11.4.6 closing note. Every numeric target in §2–§8 marked UNVERIFIED is owned by a specific Phase-0/1/2 gate or named benchmark (G1, G2, G3, G4, the soak/stress suite, the DR runbook). The verification column is the contract: an NFR is "met" only when its named test produces captured evidence (§11.4.5/§11.4.69), never when the number is merely written down here.

Sources verified

- docs/research/mvp/final/00-product-scope-and-principles.md — principles P1–P7 (§8), Mullvad-parity matrix F1–F17 + X1–X5 (§9), phase scope + exit gates G1–G6 + 8-criteria MVP DoD (§10), decisions D1–D7 incl. the iOS NE ceiling shaping D2 (§11), the "fully-responsive-web ≠ system VPN" honesty marker (§2.2).
- docs/research/mvp/final/v03-control-plane/svc-coordinator.md §7 — the convergence < 1 s SLO, helix_reconcile_seconds, revoke < 1 s, the minimal-affected-set fan-out, the 24 h no-leak soak.
- docs/research/mvp/final/v03-control-plane/svc-telemetry.md §3 — the counts-only Prometheus registry, the forbidden-label cardinality bound (no tenant_id/device_id/ip labels), the SLO→alert mapping; §10 the test matrix.
- docs/research/mvp/final/v05-security/no-logging-as-code.md — the no-logging invariant as testable NFRs (schema-lint, payload-lint, audit meta guard, ephemeral presence, coarse last_seen_at, the §11.4.108 runtime signature), the honest-boundary residual risk T5.
- docs/research/mvp/final/v02-data-plane/transport-selection-ladder.md — the escalation-ladder budgets, monotonic-cursor invariant (L-I8), and aggregate no-per-user telemetry (I5) behind NFR-006 / NFR-410.
- docs/research/mvp/final/v05-security/pki-and-certs.md — device/gateway key rotation cadence + revoke-triggered re-key behind NFR-412; the decision register D-AC-3 (audit retention, no auto-prune default) behind NFR-309.
- docs/research/mvp/final/v08-testing/ddos.md — the DDoS/flood-simulation harness behind NFR-414 (data-plane) distinct from NFR-413 (control-plane API rate limiting).

- docs/research/mvp/final/v06-deploy/codegen-pipeline.md — the D-CODEGEN-TRACK/ D-OPENAPI-AUTHORING decisions and the schema-diff breaking-change lint behind NFR-702; docs/research/mvp/final/v06-deploy/ha-and-multiregion.md and disaster-recovery.md — rolling-upgrade + RTO/RPO context behind NFR-700/701; docs/research/mvp/final/v06-deploy/observability.md — the sizing dashboard behind NFR-703.

Constitution bindings applied: §11.4.44 (revision header), §11.4.6 (no-guessing — every unproven target marked UNVERIFIED and owned by a named gate/benchmark, never asserted as measured fact), §11.4.169 (each NFR names the test type whose captured evidence proves it), §11.4.25 (coverage ledger discipline — §9 traceability), §11.4.54 (stable append-only HVPN-NFR-NNN ids), §11.4.81 (cross-platform parity — NFR-609), §11.4.108 (runtime signature — NFR-308), §11.4.65/§11.4.153 (HTML+PDF[+DOCX] exports follow in refinement).